MIL-C-27016A(USAF)
29 September 1961
Superseding
MIL-C-27016(USAF)
29 March 1960

MILITARY SPECIFICATION

CONTROL-MONITOR DCU-9/A

1. SCOPE. This specification establishes the requirements for one type of in-flight control and monitor equipment designated as Control-Monitor DCU-9/A.

2. APPLICABLE DOCUMENTS. The following documents, of the issue in effect on the date of invitation for bids, shall form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-T-9107	Test Reports, Preparation Of
MIL-E-17555	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-16	Electrical and Electronic Reference Designations
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

PUBLICATIONS

Department of Defense

Index of Specifications and Standards

FSC 1190

THE REPUBLICAN PRESS, Hamilton, Ohio
November 6, 1961 1000

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Air Force-Navy Aeronautical Bulletin

143	Specifications and Standards, Use Of

Air Force Specification Bulletin

23	Material and Process Specification

DRAWINGS

Air Force

59D46260	Control-Monitor DCU-9/A

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION. This specification makes provisions for preproduction testing.

3.2 COMPONENT PARTS. The Control-Monitor DCU-9/A includes the following components: A rotary selector switch with mechanical interlock, a push button lamp test switch, a dimmer rheostat, a press-to-test indicator light, two panel lights, a holding relay, two zener diodes, an edge-lighted plastic panel and an AN electrical connector. The components are housed in a rectangular shaped aluminum case. The electrical connector is mounted on the rear side of the case with the pins protruding to the outside and the other components are mounted on the front panel which is equipped with Dzus fasteners for panel mounting. The case is provided with a removable adapter mount which is used for mounting to the T-19 Shock mount rack assembly as an additional mounting capability.

3.3 GENERAL SPECIFICATION. The requirements of MIL-E-5400 are applicable as requirements of this specification. Where the requirements of MIL-E-5400 and this specification conflict, the requirements of this specification shall govern.

3.4 SELECTION OF SPECIFICATIONS AND STANDARDS. Specifications and standards for all materials, parts and processes, which are not specifically designated herein and which are necessary for the execution of the Statement of Work shall be selected in the following manner:

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3.4.1 ORDER OF PRECEDENCE. Specifications and standards shall be selected in accordance with Group I of ANA Bulletin No. 143. This selection shall first be made from USAF Specification Bulletin No. 23 for materials and processes. In the event that the necessary documents cannot be found in USAF Bulletin No. 23, selection shall be made from the Department of Defense Index of Specifications and Standards. Should an applicable Government specification not exist, selection shall be made from Group II and Group III, of ANA Bulletin No. 143, in the order named. In the event a document under Group III is required for proper execution of the Statement of Work, the contractor shall obtain approval to use it from the procuring activity.

3.5 STANDARD PARTS. MS or AN Standard parts shall be identified by their complete part numbers. Commercial parts may be used in applications for which no suitable corresponding MS or AN part is in effect on the date of request for proposals or is not available from any of the suppliers who stock MS and AN parts, provided they conform to all requirements of the procurement activity. In this case, the commercial part must be replaceable by the standard part (MS or AN) without alteration, and the corresponding standard part number must be referenced in the parts list. The use of socket head screws shall be avoided when frequent operational adjustment or removal is required.

3.6 PROTECTIVE TREATMENT. When materials are used in the construction of the Control-Monitor DCU-9/A that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.7 DESIGN. The design and construction of the Control-Monitor DCU-9/A shall be in accordance with drawing 59D46260 and all subsidiary detail drawings. The electrical wiring shall be dressed and maintained in such a position that no interference occurs between the assembly and the case. The clearance between current carrying parts and the case shall be such that the requirements of 4.5.3 are fulfilled.

3.8 PERFORMANCE. The DCU-9/A shall satisfactorily perform specific in-flight control and monitoring functions in connection with certain special stores carried in aircraft.

3.8.1 EQUIPMENT OPERATING. The DCU-9/A shall be capable of satisfactory operation, as specified in 3.8, while being subjected to, and after subjection to, the following service conditions:

a. VIBRATION. Vibration coincident to handling, shipment and use.

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b. HIGH TEMPERATURE. Temperatures up to and including 71°C (160°F) with a relative humidity of not more than 5 percent.

c. LOW TEMPERATURE. A low temperature of -54°C (-65°F).

d. HUMIDITY. Conditions of relative humidity ranging up to 95 percent and above for both continuous and intermittent periods, including conditions wherein condensation takes place in and on the equipment.

e. ALTITUDE. An altitude of 50,000 feet above sea level at a temperature of -54°C (-65°F).

3.8.2 EQUIPMENT NON-OPERATING. The DCU-9/A shall be capable of satisfactory operation as specified in 3.8 after being subjected to the following conditions:

a. SHOCK. Shock coincident to handling, shipment and use.

b. TEMPERATURE SHOCK. Extreme variations of temperature, from 85°C (185°F) to -40°C (-40°F).

c. FUNGUS. Prolonged exposure to tropical atmospheres containing fungus spores.

d. SALT SPRAY. Prolonged exposure to salt spray such as generally encountered in an area near a large body of salt water.

e. SAND AND DUST. Prolonged exposure to desert atmospheres containing sand and dust particles.

3.9 INTERCHANGEABILITY. Parts functionally and dimensionally interchangeable shall be identified by the same part number. When parts are not interchangeable, the criteria listed in MIL-D-70327 shall apply and new drawings shall be prepared as required.

3.10 IDENTIFICATION OF PRODUCT. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10.1 REFERENCE DESIGNATIONS. Reference designations used for marking within the unit shall be in accordance with MIL-STD-16. Markings shall not adversely affect the leakage path between conductors, or any other factor of performance. Reference symbols shall be employed to identify each part for its particular circuit application. The same reference symbol shall be used for identifying a particular part on all types of data, including drawings, diagrams, and spare parts lists prepared for the equipment. When practicable, markings shall be placed on the supporting structure adjacent to each replaceable part in such a position as to be readily viewed.

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3.11 WORKMANSHIP. The DCU-9/A, including all parts and accessories, shall be fabricated and finished in a workmanlike manner to produce units free of all defects. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, machine screw assemblage and freedom from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS. The inspection and testing of the DCU-9/A shall be as follows:

a. Preproduction Tests

b. Acceptance Tests

4.2 PREPRODUCTION TESTING

4.2.1 PREPRODUCTION TEST SAMPLES. The preproduction test samples shall consist of models representative of the production items. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant, under the supervision of the procuring activity.

4.2.1.1 In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report according to MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

4.2.2 PREPRODUCTION TESTS. Preproduction tests shall consist of all tests described herein under "4.5 TEST METHODS."

4.3 ACCEPTANCE TESTS. Acceptance tests shall consist of:

a. Individual Tests

b. Sampling Plan and Tests

4.3.1 INDIVIDUAL TESTS. Each DCU-9/A shall be subjected to the following tests as described under "4.5 TEST METHODS" of this specification:

a. Examination of Product

b. Electrical Performance Test

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4.3.2 SAMPLING PLAN AND TESTS

4.3.2.1 SAMPLING PLAN. Samples shall be selected at random in accordance with the following schedule, and subjected to the tests listed below and described under "4.5 TEST METHODS" of this specification:

Consecutively Produced Units on Order	Number of Samples
99 or less	2
100 to 299	1
Each 200 or fraction thereof on balance of order	1 unit for each additional 200 produced

a. Vibration Test

b. Shock Test

4.3.2.1.1 REJECTION AND RETEST. When one or more units from a production run fails to meet the specified tests, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.3.2.1.2 INDIVIDUAL TESTS MAY CONTINUE. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of units on hand or produced later shall not be made until it is determined that they meet all the requirements of this specification.

4.3.3 DEFECTS IN ITEMS ALREADY ACCEPTED. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 TEST CONDITIONS

4.4.1 ATMOSPHERIC CONDITIONS. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 25°C ± 10 C and a relative humidity of 80 percent or less. When tests are made at atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.5 TEST METHODS

4.5.1 EXAMINATION OF PRODUCT. Each DCU-9/A shall be examined to determine compliance with the requirements of this specification for

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which tests are not specified including workmanship, materials, ease of operation of controls, and strict conformance to the pertinent drawings.

4.5.2 ELECTRICAL PERFORMANCE TEST. An electrical performance test shall be conducted on each DCU-9/A to insure continuity of all electrical circuits and to insure that no breakdown of any circuits or components shall occur due to the application of high voltages. This test shall be conducted in four parts. Part "A" will be the Continuity Test; Part "B", the Hi-Pot Test; Part "C", the Diode Test; and Part "D", the Relay Test.

4.5.2.1 TEST CIRCUIT. The schematic diagram of the acceptance tester circuit is shown in figure 1.

4.5.2.2 TEST CONDITION. The DCU-9/A shall be tested when completely assembled in its case and all mounting screws fully tightened. The connector mounting screw around which the case has been spotfaced shall be used for the ground connection to the DCU-9/A.

4.5.2.3 ELECTRICAL TEST PROCEDURE. The test procedure shall be in accordance with Table I.

4.5.2.4 REJECTION CRITERIA. Failure to obtain the prescribed indication during any step of the test procedure in table I shall be cause for rejection of the Control-Monitor DCU-9/A.

4.5.3 VIBRATION TEST. The vibration test shall be in accordance with the vibration tests, procedure XII of MIL-E-5272. The high temperature portion of this test will be conducted at 71°C (160°F). Upon completion of the vibration test, the equipment shall be examined for any physical damage and tested in accordance with the Electrical Performance Test, (4.5.2) of this specification.

4.5.4 SHOCK TEST. The shock test shall be in accordance with the shock tests, procedure V of MIL-E-5272. Upon completion of the shock tests, the equipment shall be examined and tested in accordance with 4.5.2.

4.5.5 HIGH TEMPERATURE TEST. The high temperature test shall be in accordance with the high temperature tests, procedure II of MIL-E-5272. At the termination of 25 and 50 hours, while at this temperature, the equipment shall be examined and operated in accordance with 4.5.2.

4.5.6 LOW TEMPERATURE TEST. The low temperature test shall be in accordance with the low temperature test, procedure I of MIL-E-5272. At the conclusion of the exposure period and while at this temperature, the equipment shall be examined and tested in accordance with 4.5.2.

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-NOTES-

1- AN CONNECTOR SPECIFIED BELOW IS CABLE TYPE TO PERMIT USE OF A FLEXIBLE CABLE BETWEEN DCU-9/A (OR MOD KIT ASSY) & TEST UNIT

2- CONTACTS OF RELAYS K1 THRU K16 MUST BE RATED TO BREAK 1000 AC AT 100 MA

3- CONTACTS OF RELAYS K17 AND K19 MUST WITHSTAND 1000 VAC GROUND TEST

4- T1 MUST WITHSTAND 2500 VAC GROUND TEST. T2 MUST WITHSTAND 500 VAC GROUND TEST.

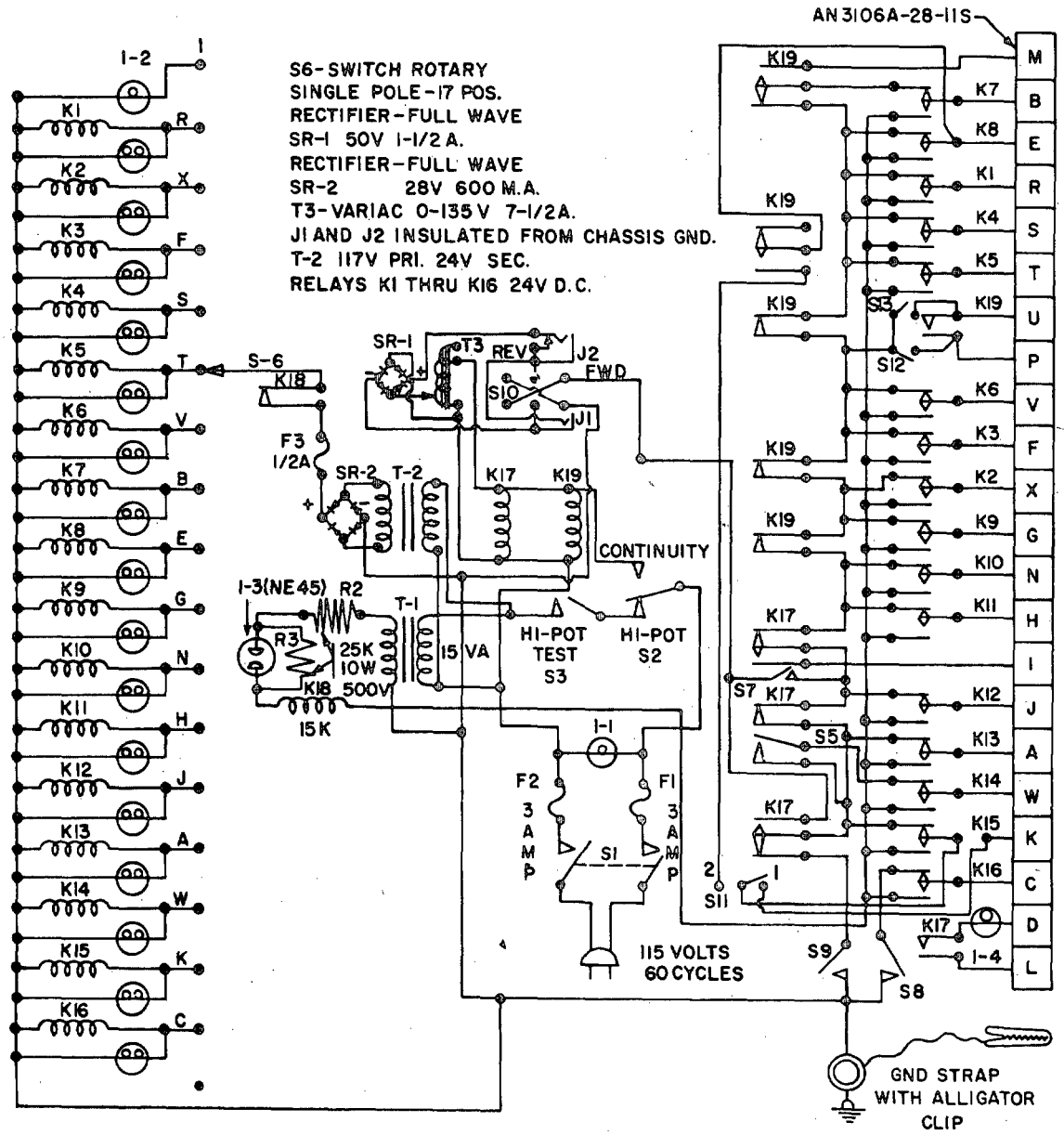


FIGURE 1 ACCEPTANCE TESTER SCHEMATIC

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TABLE I

CONTROL-MONITOR DCU-9/A	ACCEPTANCE TESTER	INDICATION
PART A - CONTINUITY TEST		
	PREPARATION FOR TEST	
1. Bulbs in Warning and Panel Light Sockets	1. Power Switch in "OFF" position	
2. Rotary Switch S-1 in "OFF" position	2. Voltage Control, T-3, in "Zero" position	
3. "DIM" rheostat in extreme counterclockwise or "OFF" position	3. 0-50V DC Voltmeter in J-1	
	4. Continuity/Hi-Pot Switch, S-2, in "CONTINUITY" position	
	5. Hi-Pot Switch, S-3, in "OFF" position	
	6. Switches S-5, S-6, S-7, S-8 and S-9 in "OFF" position	
	7. Switch, S-11, in position 1	
	8. Switch, S-10, in "Forward" position	
	9. AN connector connected to the DCU-9/A and ground clip attached to grounding screw on DCU-9/A referenced in 4.5.2.2	

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CONTROL-MONITOR DCU-9/A	ACCEPTANCE TESTER	INDICATION
PART A - CONTINUITY TEST		
	PREPARATION FOR TEST	
	10. Power plug inserted in 110 volts AC receptacle	
	PERFORMANCE OF TEST	
	1. Place Switch, S-1, in "ON" position	1. Lamp, I-1, on tester shall light
	2. Operate, T-3, to increase voltage	2. DC Voltmeter shall read 28 volts. Lamp, I-4, on tester shall light
3. Operate "DIM" rheostat control fully clockwise and slowly back to full counterclockwise position		3. All panel lettering except "DCU-9/A" shall become clearly illuminated under subdued lighting conditions when the control is in the full clockwise position. As the control is rotated counterclockwise the illumination shall decrease, finally going out when the control is in the fully counterclockwise position.
	4. Place Switch, S-5, in "ON"	4. All panel lettering except "DCU-9/A" shall be brightly illuminated under subdued lighting conditions. In this test the "DIM" con-

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CONTROL-MONITOR DCU-9/A	ACCEPTANCE TESTER	INDICATION
PART A - CONTINUITY TEST	PERFORMANCE OF TEST	
		trol has no effect upon intensity of panel light illumination.
5. Place Rotary Switch, S-1, in "SAFE" position.		5. N/A
6. Press lamp test switch, S-2		6. Warning lamp, I-1, on DCU-9/A shall light.
	7. Place Switch, S-8, in "ON" position.	7. Warning lamp, I-1, shall light and remain lit.
8. Press the "Press-to-Test" warning lamp, I-1		8. The warning lamp, I-1, shall go out momentarily during "press-to-test" operation and shall light again upon complete depression of the lamp.
9. Place rotary switch lock in "SGA" position. Then operate rotary switch, S-1, to the "GND" position.		9. The warning lamp, I-1, shall go out momentarily during switching operation and shall light again when switch, S-1, reaches "GND" position.

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CONTROL-MONITOR DCU-9/A	ACCEPTANCE TESTER	INDICATION
PART A - CONTINUITY TEST		
	PERFORMANCE OF TEST	
10. Rotate switch, S-1, to "AIR" position		10. The warning lamp, I-1, shall go out momentarily during switching operation and shall light again when switch, S-1, reaches "AIR" position.
11. Return switch, S-1, to "SAFE" position. Return switch lock to "OS" position. Return switch, S-1, to "OFF" position.		11. N/A
	12. Place switch, S-8, in "OFF" position	12. Warning lamp, I-1, on DCU-9/A shall go out.
	13. Place switch, S-8, in "ON" position	13. Warning lamp, I-1, shall remain out.
	14. Place switch, S-7, in "ON" position	14. Warning lamp, I-1, shall light.
	15. Place switch, S-7, in "OFF" position	15. Warning lamp, I-1, shall remain lit as result of K-2 relay action.
	16. Return DC Voltage Control T-3 to "ZERO" position	16. N/A
	17. Place switch, S-1, in "OFF" position	17. N/A

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CONTROL-MONITOR DCU-9/A	ACCEPTANCE TESTER	INDICATION
PART B - HI-POT TEST		
	PREPARATION FOR TEST	
1. Bulbs removed from warning and panel light sockets.	1. Power switch, S-1, in "OFF" position.	
2. Switch, S-1, in "OFF" position.	2. Continuity/Hi-pot switch, S-2, in "Hi-pot" position.	
3. "DIM" rheostat control in extreme counter-clockwise position.	3. AN connector connected to the DCU-9/A and ground clip attached to grounding screw on DCU-9/A referenced in para. 4.5.2.2.	
	4. Switches, S-5, S-6, S-7, and S-8 in "OFF" position.	
	5. Switch, S-9, in "ON" position	
	6. Switch, S-11, in position 1	
	7. Power input plug inserted in 110 volt AC receptacle	
	PERFORMANCE OF TEST	
	1. Place power switch, S-1, in "ON" position	1. Lamp, I-1, on tester shall light
	2. Place switch, S-6, in position 1	2. N/A

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CONTROL-MONITOR DCU-9/A	ACCEPTANCE TESTER	INDICATION
PART B - HI-POT TEST	PERFORMANCE OF TEST	
	3. Place Hi-pot switch, S-3, in "ON" position	3. Lamp, I-2, on tester shall light
	4. Operate switch, S-6, thru positions R to K	4. Lamps R thru K on tester shall light in sequence
	5. Operate switch, S-6, to position C	5. Lamp, I-3, on tester shall light
	6. Place switch, S-9, in "OFF" position	6. N/A
7. Operate rotary switch, S-1, to "SAFE" position. Operate switch lock to "SGA" position. Operate rotary switch, S-1, to "GND" position.		7. N/A
	8. Operate switch S-6 thru positions R to C.	8. Lamps R thru C on tester shall light in sequence. Lamp, I-3, on tester shall not light.
	9. Place switch S-9 in "ON" position	9. N/A
	10. Operate switch S-6 thru positions R to C.	10. Lamps R thru C on tester except for J, K, and C shall light in sequence. Lamp, I-3, on tester shall light for positions J, K, and C.

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CONTROL-MONITOR DCU-9/A	ACCEPTANCE TESTER	INDICATION
PART B - HI-POT TEST		
	PERFORMANCE OF TEST	
	11. Place switch S-12 in "ON position	11. N/A
12. Place switch S-1 in "OFF" position		12. N/A
	13. Operate switch S-6 thru positions R thru C	13. Lamps R thru C on tester shall light in sequence. Lamp I-3 on tester shall light on V and C
14. Place switch S-1 in "SAFE" position		14. N/A
	15. Operate switch S-6 thru positions R thru C	15. Lamps R thru C on tester shall light in sequence. Lamp I-3 on tester shall light on V, J, K and C.
16. Place rotary switch lock in "SGA" position then place switch S-1 in "GND"		16. N/A
	17. Operate switch S-6 thru positions R thru C	17. Lamps R thru C on tester shall light in sequence. Lamp I-3 on tester shall light on X, J, K and C.
18. Place switch S-1 in "AIR" position		18. N/A

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CONTROL-MONITOR DCU-9/A	ACCEPTANCE TESTER	INDICATION
PART B - HI-POT TEST		
	PERFORMANCE OF TEST	
	19. Operate switch S-6 thru positions R thru C	19. Lamps R thru C on tester shall light in sequence. Lamps I-3 on tester shall light on N, J, K and C.
	20. Place S-12 in "OFF" position	20. N/A
	21. Place switch S-13 in "ON" position	21. N/A
22. Switch S-1 in "AIR" position		22. N/A
	23. Operate switch S-6 thru positions R thru C	23. Lamps R thru C on tester shall light in sequence. Lamp I-3 on tester shall light on T, J, K and C.
24. Place switch S-1 in "GND" position		24. N/A
	25. Operate switch S-6 thru positions R thru C	25. Lamps R thru C on tester shall light in sequence. Lamp I-3 on tester shall light on S, J, K and C.
26. Place switch S-1 in "SAFE" position		26. N/A
	27. Operate switch S-6 thru positions R thru C	27. Lamps R thru C on tester shall light in sequence. Lamp I-3 on tester shall light on R, J, K and C.
28. Place rotary switch in "OS" position then place switch S-1 in "OFF" position		28. N/A

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CONTROL-MONITOR DCU-9/A	ACCEPTANCE TESTER	INDICATION
PART B - HI-POT TEST		
	PERFORMANCE OF TEST	
	29. Operate switch S-6 thru positions R thru C	29. Lamps R thru C on tester shall light in sequence. Lamp I-3 on tester shall light on R and C.
PART C - DIODE TEST		
	PREPARATION FOR TEST	
1. Bulbs removed from warning and panel light sockets.	1. Power switch, S-1, in "OFF" position	
2. Rotary Switch, S-1, in "SAFE" position	2. DC Voltage Control, T-3, to "ZERO" position	
3. "DIM" rheostat control in extreme counter-clockwise position.	3. Continuity/Hi-Pot switch in "CONTINUITY" position.	
	4. Hi-pot switch, S-3, in "OFF" position	
	5. Bulb of lamp, I-4, removed from socket	
	6. Switch, S-8, in "ON" position	
	7. DC Voltmeter connected to J-1	

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CONTROL-MONITOR DCU-9/A	ACCEPTANCE TESTER	INDICATION
PART C - DIODE TEST		
	PREPARATION FOR TEST	
	8. 0-100 MA meter connected to J-2	
	9. Switch, S-10, in "FORWARD" position	
	10. Switch, S-11, in position 1	
	11. Switches, S-5, S-7 and S-9 in "OFF" position	
	12. Tester connected to DCU-9/A and power source as indicated in Part A	
	PERFORMANCE OF TEST	
	1. Place power switch, S-1, in "ON" position	1. N/A
	2. Adjust Voltage Control "T-3" to obtain 14 Volts	2. DC Voltmeter shall read 14. Note reading obtained on MA meter in "FORWARD" position of S-10.

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CONTROL-MONITOR DCU-9/A	ACCEPTANCE TESTER	INDICATION
PART C - DIODE TEST		
	PERFORMANCE OF TEST	
	3. Place switch, S-10, in "REVERSE" position.	3. Note reading obtained on MA meter in "REVERSE" position of S-10. Normal current readings in steps 2 and 3 shall be in the range of 30-40 ma and shall not differ by more than 2 ma. If current readings differ by more than 2 ma the diode is defective which was conducting in the "reverse" direction when the higher reading was obtained.
	4. Remove MA meter from J-2.	4. N/A
PART D - RELAY TEST		
	PREPARATION FOR TEST	
1. Bulbs in warning and panel light sockets.	1. Power switch, S-1, in "OFF" position.	
2. Rotary switch, S-1, in "OFF" position.	2. Bulb of lamp, I-4, in socket.	

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CONTROL-MONITOR DCU-9/A	ACCEPTANCE TESTER	INDICATION
PART D - RELAY TEST		
	PREPARATION FOR TEST	
3. "DIM" rheostat control in extreme counter-clockwise position.	3. Voltage control, T-3, in "ZERO" position. 4. Switch, S-3, in "OFF" position. 5. Switch, S-2, in "CONTINUITY" position. 6. Switch, S-8, in "ON" position. 7. Switch, S-9, in "OFF" position. 8. Switch, S-11, in position 2. 9. Switches, S-5, and S-7, in "OFF" positions 10. Voltmeter, 0-30 V DC, connected to J-1. 11. Tester connected to DCU-9/A and power source as indicated in Part A.	
	PERFORMANCE OF TEST	
	1. Place switch, S-1, in "ON" position.	1. Lamp, I-1, on tester shall light.

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CONTROL-MONITOR DCU-9/A	ACCEPTANCE TESTER	INDICATION
PART D - RELAY TEST	PERFORMANCE OF TEST	
	2. Operate control, T-3, to slowly increase DC voltage until lamp, I-4, on tester lights.	2. Voltmeter reading shall not exceed 22.5 volts.
	3. Operate control, T-3, to slowly decrease DC voltage until lamp, I-4, on tester goes out.	3. Voltmeter reading shall not be less than 4.3 volts.
	4. Place switch, S-1, in "OFF" position.	4. Lamp, I-1, on tester shall go out.
	5. Disconnect DCU-9/A from Acceptance Tester.	5. N/A

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4.5.7 TEMPERATURE SHOCK TEST. The temperature shock test shall be in accordance with the temperature shock tests, procedure I of MIL-E-5272. At the conclusion of the three temperature cycles, the equipment shall be returned to room temperature, examined and tested in accordance with 4.5.2.

4.5.8 HUMIDITY TEST. The humidity test shall be in accordance with the humidity tests, procedure I of MIL-E-5272. During every fifth cycle, one hour prior to the conclusion of the six hour temperature period, the equipment shall be operated and tested in accordance with 4.5.2. At the conclusion of the 240 hour period, the equipment shall be returned to the prevailing room conditions, examined and tested in accordance with 4.5.2. When conducting the Electrical Performance Test, only 250 volts shall be applied to the dimmer rheostat in lieu of 500 volts.

4.5.9 FUNGUS TEST. The fungus test shall be in accordance with the fungus resistance tests, procedure I of MIL-E-5272. The unit selected for this test shall not have previously been subjected to the salt spray test or the sand and dust test. Upon completion of the test, the equipment shall be thoroughly examined to ascertain that there is no evidence of fungus growth which would impair the function of the equipment. The equipment shall then be tested in accordance with 4.5.2.

4.5.10 ALTITUDE TEST. The altitude test shall be in accordance with the altitude tests, procedure VI, condition C of MIL-E-5272 at an altitude of 50,000 feet. The ambient temperature within the test chamber shall be reduced to -54°C (-65°F). While under these conditions, the equipment shall be tested in accordance with 4.5.2. The equipment shall be returned to the prevailing room conditions, examined and tested in accordance with 4.5.2.

4.5.11 SALT SPRAY TEST. The salt spray test shall be in accordance with the salt spray test of MIL-E-5272, except that the test shall be of 50 hours duration. At the conclusion of this test period, the salt residue shall be removed and the equipment examined and tested in accordance with 4.5.2. When conducting the Electrical Performance Test, only 250 volts shall be applied to the dimmer rheostat in lieu of 500 volts.

4.5.12 SAND AND DUST TEST. The sand and dust test shall be in accordance with the sand and dust tests, procedure I of MIL-E-5272, except that velocity of sand and dust through the chamber shall be 2500 $\pm$ 500 feet per minute. Upon completion of the test, the equipment shall be examined and tested in accordance with 4.5.2.

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5. PREPARATION FOR DELIVERY

5.1 PRESERVATION, PACKING AND PACKAGING. Preservation, packing and packaging for shipment shall be in accordance with method II of MIL-E-17555. Levels of protection shall be as specified by the procuring activity.

5.2 MARKING FOR SHIPMENT AND STORAGE. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be:

CONTROL-MONITOR DCU-9/A

6. NOTES

6.1 INTENDED USE. The DCU-9/A is intended for use as a replacement for the T-249 In-Flight Control Tester. During the interim period the T-249 and DCU-9/A may be used interchangeably.

6.2 ORDERING DATA. Invitations for bids, contracts, and purchase order should specify the following:

a. Title, number and date of this specification.

b. Whether the preproduction tests will be conducted by a laboratory designated by the procuring activity or by the contractor under the supervision of the procuring activity.

c. Required levels of preservation, packaging and packing.

6.3 PREPRODUCTION TEST SAMPLES. It is expected that the contract or purchase order will specify that a minimum of two DCU-9/A's will be required as preproduction test samples.

6.4 SOURCES. When requesting publications, refer to both title and number.

6.5 DEFINITIONS

a. PREPRODUCTION TESTS. Preproduction tests are those tests performed prior to production on representative models of the production equipment to demonstrate that the equipment meets the requirements of this specification.

b. ACCEPTANCE TESTS. Acceptance tests are those tests accomplished on equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the units are equivalent in performance and construction to the approved preproduction samples.

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